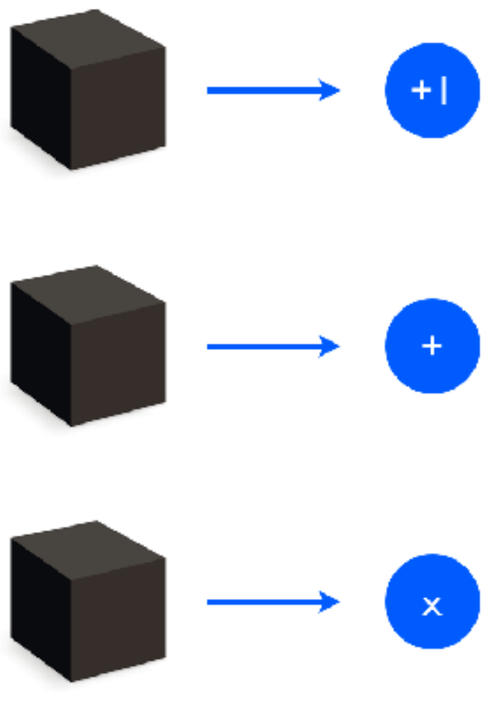
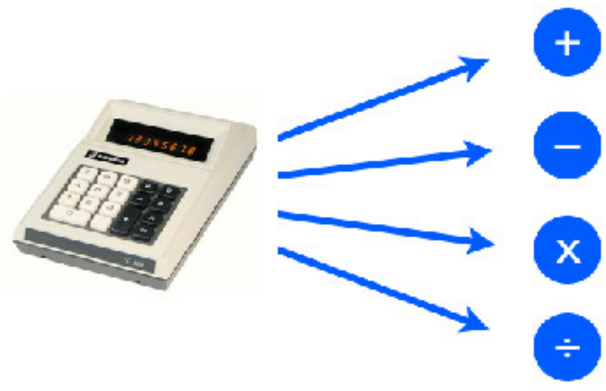
36. Machines and Programs

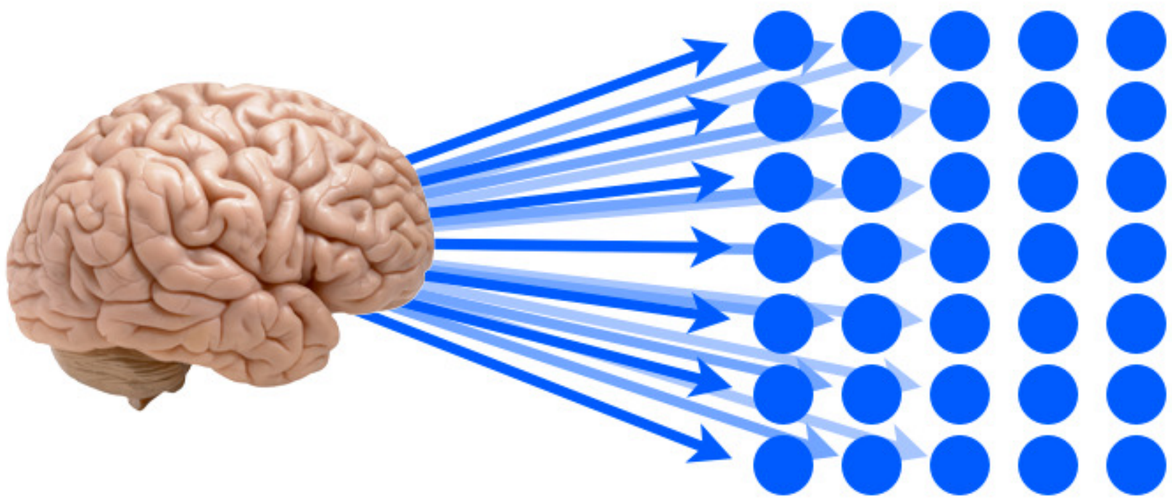
Multi-function Machines



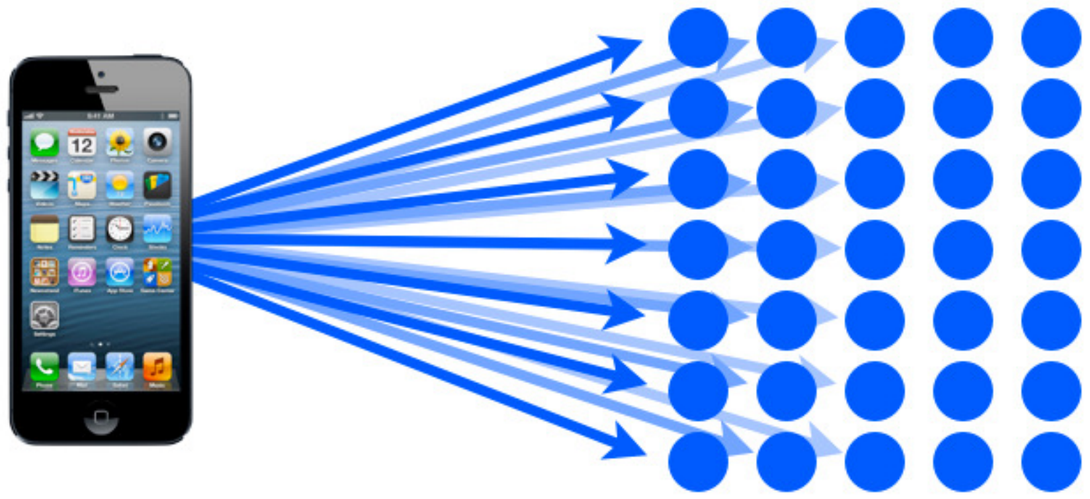
So far, all of the machines we have encountered compute one function each. Of course, it is possible to construct a machine which computes more than one function. A simple calculator computes four functions.



Our brains can compute a lot more than four functions. In fact, it seems like there is no practical limit on the number of functions a brain could compute. It is an *infinite* function machine.

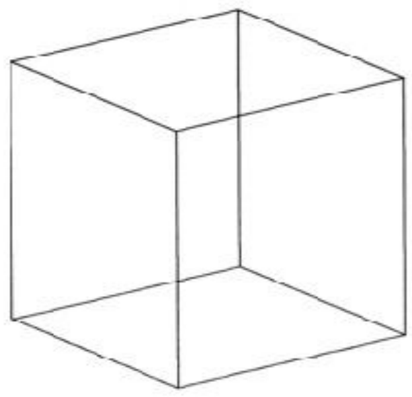


This isn't just true of our brains. It's true of any computer which can read *programs*. Such devices are known as **universal computers**. Your laptop and your phone are universal computers.



Universal Turing Machines: The Goal

The goal: Create a Turing Machine which can perform the task of any other Turing Machine. This is called a **Universal Turing Machine** (UTM).



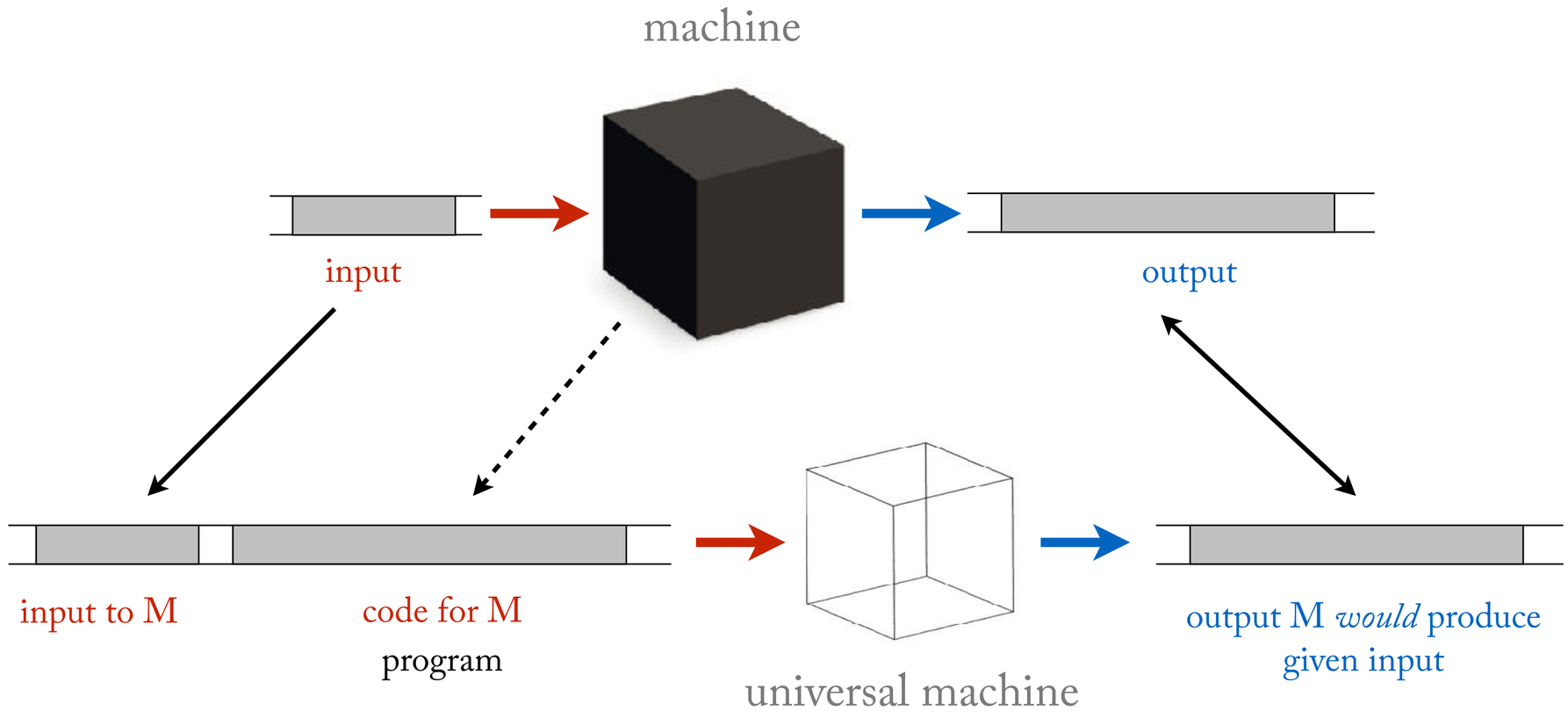
universal machine

Universal Turing Machines: The Idea

Strategy for designing a UTM:

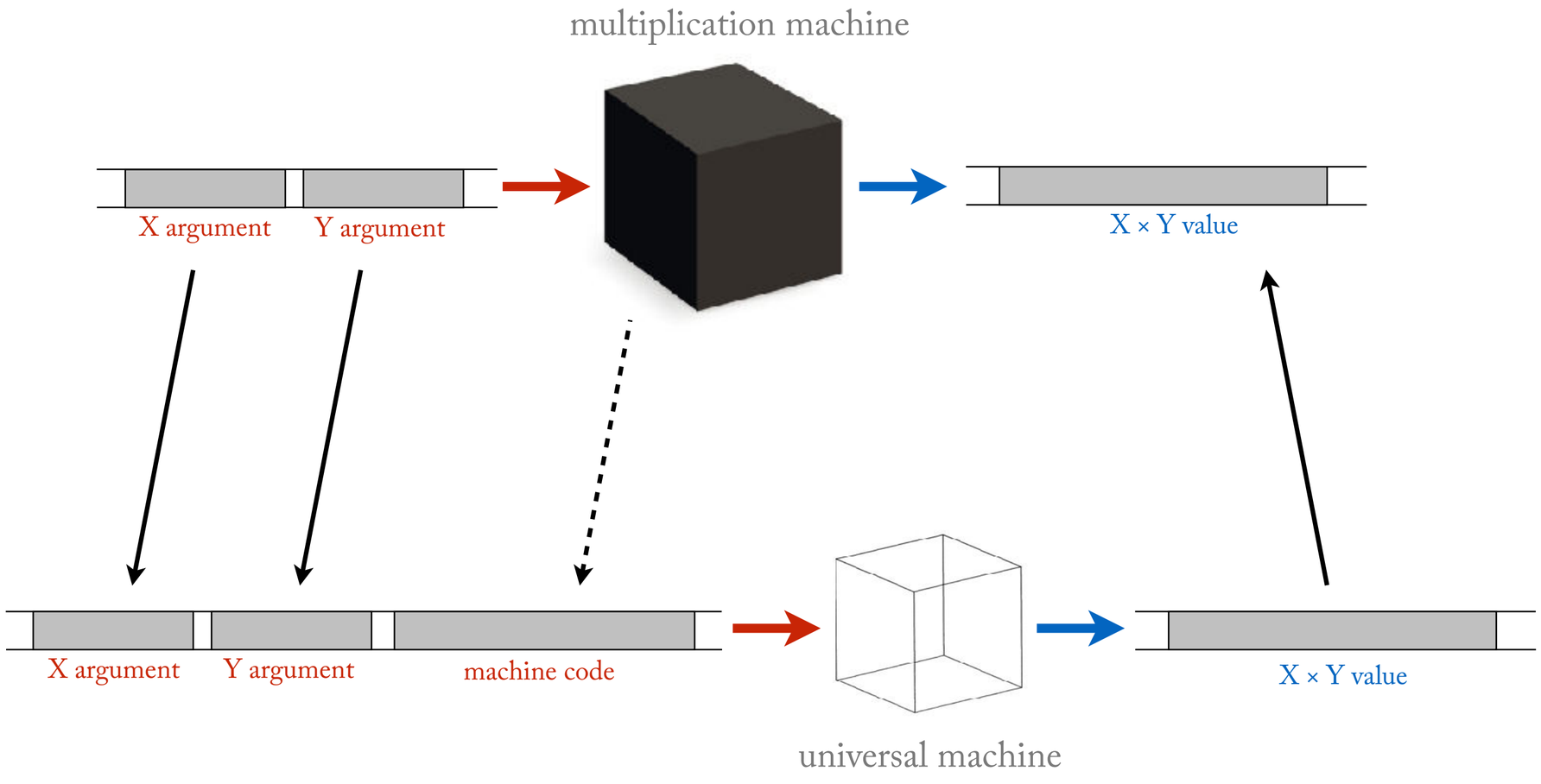
Stage 1: find a method for turning an arbitrary Turing Machine (the simulated machine) into something a Turing Machine can read— i.e. 1's and 0's on tape. This something is called the simulated machine's **code number**.

Stage 2: invent a UTM which takes as input (i) any simulated machine's code number, and (ii) the simulated machine's input; the UTM then outputs the simulated machine's output.



For example, let M be a multiplication machine. It takes two inputs (corresponding to the X and Y arguments) and returns one output (corresponding to the value X×Y). And let's say that M's code number is C.

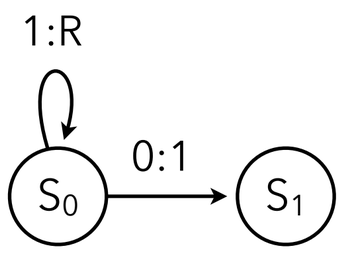
Then if we give a tape which contains, X, Y, *and* C to the UTM, it will return the value X×Y.



Codes Numbers (aka Programs)

Derive a machine's code number in five easy steps.

Step 1. Choose a Turing Machine, and draw up it's state table.



STATE	IN	OUT	NEXT STATE
S_0	1	R	S_0
	0	1	S_1
S_1	1	H	
	0	H	

Step 2. Divide each state cell into rows according to input. Eliminate rows which do not lead to another state.

STATE	IN	OUT	NEXT STATE
S_0	1	R	S_0
S_0	0	1	S_1
S_1	1	H	
S_1	0	H	

Step 3. Replace all symbols with numbers, according to a translation system.

State S_n	⟿	*n*
No state	⟿	0
Read a "0"	⟿	0
Read a "1"	⟿	1
Write a "0"	⟿	0
Write a "1"	⟿	1
Move left (L)	⟿	2
Move right (R)	⟿	3
Halt (H)	⟿	4

STATE	IN	OUT	NEXT STATE
0	1	3	0
0	0	1	1
1	1	4	0
1	0	4	0

Step 4. Represent the table by a sequence of quadruples:

state, in, out, next
state, in, out, next
etc...

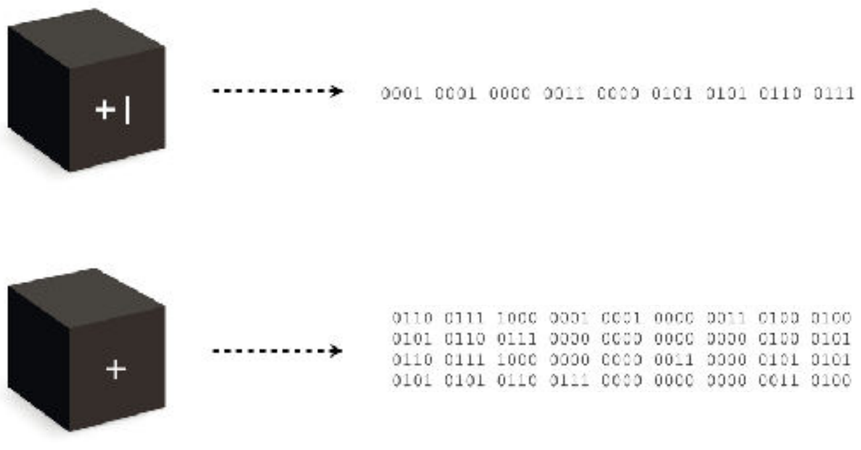
0, 1, 3, 0
0, 0, 1, 1
1, 1, 4, 0
1, 0, 4, 0

Step 5. Represent each number in tally. Represent commas with 0's and line breaks with *s. The result is a string 0's and 1's.

001011100*000010
1*10101111100*100
0111100

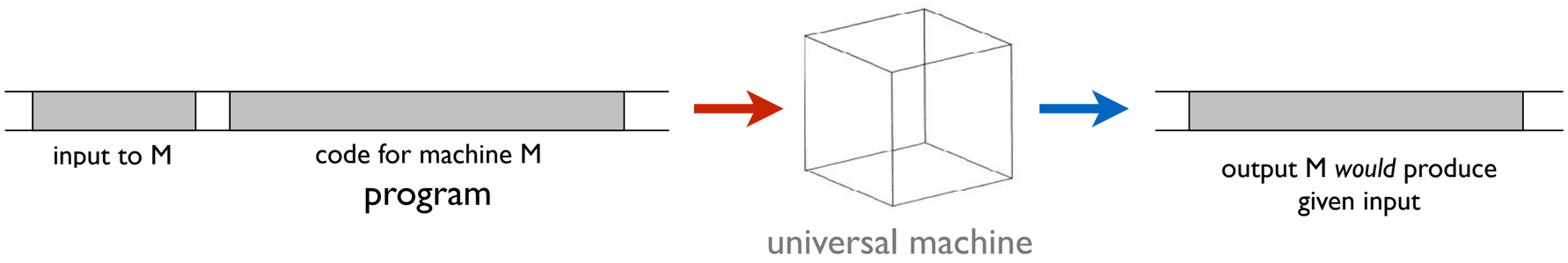
Now every single Turing Machine can be associated with a code number. And every single (well-formed) code number corresponds to a Turing Machine.

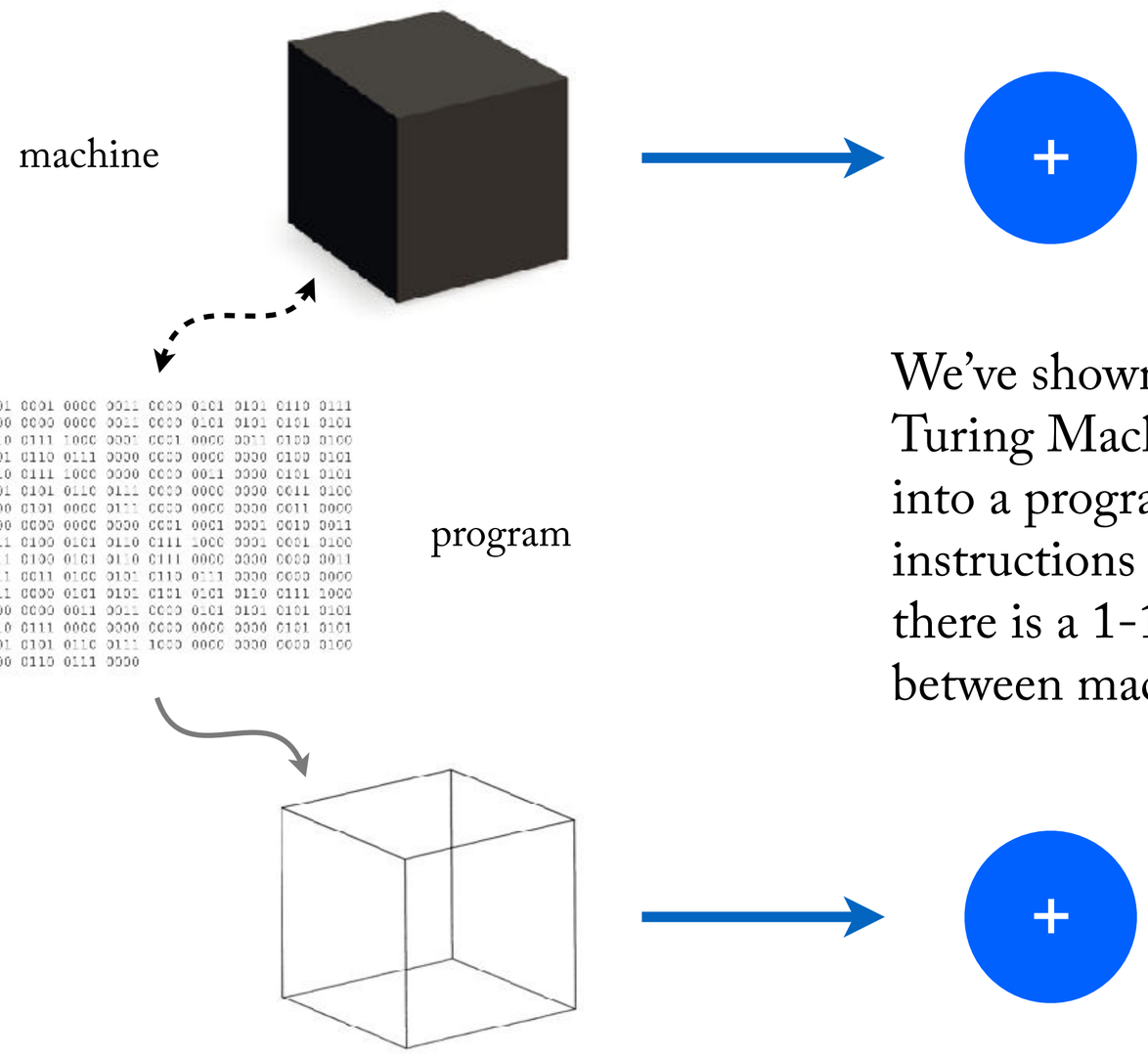
These code numbers will now become the programs which are given to the Universal Machine as input.



Machines as Programs

The code for a machine is also known as its **program**. A program is simply an algorithm written down in a language that a universal machine can interpret.





We've shown that any given Turing Machine can be turned into a program which serves as instructions for the UTM. So there is a 1-1 correspondence between machines and programs.

Up until now algorithms have been used to describe machines at a higher level of abstraction. Now they are used as something which is inputted *into* the machine. The universal machine, in effect, *reads* higher levels of abstraction.